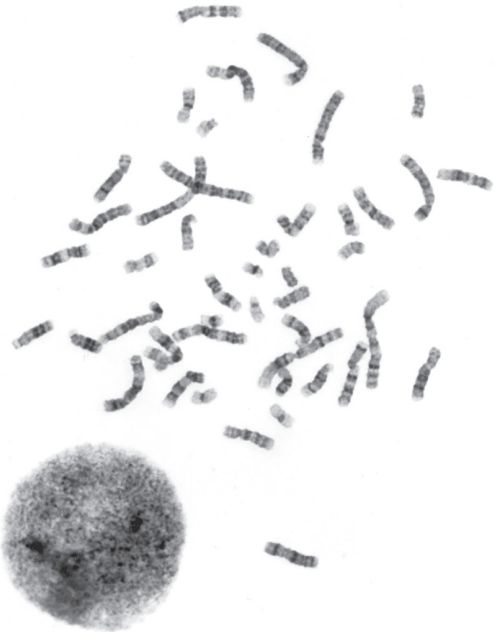


Figure 2-10 A chromosome spread prepared from a lymphocyte culture that has been stained by the Giemsa-banding (G-banding) technique. The darkly stained nucleus adjacent to the chromosomes is from a different cell in interphase, when chromosomal material is diffuse throughout the nucleus. *See Sources & Acknowledgments.*

As stated earlier, there are 24 different types of human chromosome, each of which can be distinguished cytologically by a combination of overall length, location of the centromere, and sequence content, the latter reflected by various staining methods. The centromere is apparent as a **primary constriction,** a narrowing or pinching-in of the sister chromatids due to formation of the kinetochore. This is a recognizable cytogenetic landmark, dividing the chromosome into two **arms,** a short arm designated **p** (for *petit*) and a long arm designated **q**.

Figure 2-10 shows a prometaphase cell in which the chromosomes have been stained by the Giemsa-staining (**G-banding**) method (also see Chapter 5). Each chromosome pair stains in a characteristic pattern of alternating light and dark bands (G bands) that correlates roughly with features of the underlying DNA sequence, such as base composition (i.e., the percentage of base pairs that are GC or AT) and the distribution of repetitive DNA elements. With such banding techniques, all of the chromosomes can be individually distinguished, and the nature of many structural or numerical abnormalities can be determined, as we examine in greater detail in Chapters 5 and 6.

Although experts can often analyze metaphase chromosomes directly under the microscope, a common procedure is to cut out the chromosomes from a digital image or photomicrograph and arrange them in pairs in a standard classification (Fig. 2-11). The completed picture is called a **karyotype**. The word *karyotype* is also used to refer to the standard chromosome set of an individual ("a normal male karyotype") or of a species ("the human karyotype") and, as a verb, to the process of preparing such a standard figure ("to karyotype").

Unlike the chromosomes seen in stained preparations under the microscope or in photographs, the chromosomes of living cells are fluid and dynamic structures. During mitosis, the chromatin of each interphase chromosome condenses substantially (Fig. 2-12). When maximally condensed at metaphase, DNA in chromosomes is approximately 1/10,000 of its fully extended state. When chromosomes are prepared to reveal bands (as in Figs. 2-10 and 2-11), as many as 1000 or more bands can be recognized in stained preparations of all the chromosomes. Each cytogenetic band therefore contains as many as 50 or more genes, although the density of genes in the genome, as mentioned previously, is variable.

Meiosis

Meiosis, the process by which diploid cells give rise to haploid gametes, involves a type of cell division that is unique to germ cells. In contrast to mitosis, meiosis consists of one round of DNA replication followed by *two* rounds of chromosome segregation and cell division (see meiosis I and meiosis II in Fig. 2-13). As outlined here and illustrated in Figure 2-14, the overall sequence of events in male and female meiosis is the same; however, the timing of gametogenesis is very different in the two sexes, as we will describe more fully later in this chapter.

Meiosis I is also known as the **reduction division** because it is the division in which the chromosome number is reduced by half through the pairing of homologues in prophase and by their segregation to different cells at anaphase of meiosis I. Meiosis I is also notable because it is the stage at which genetic **recombination** (also called **meiotic crossing over**) occurs. In this process, as shown for one pair of chromosomes in Figure 2-14, homologous segments of DNA are exchanged between nonsister chromatids of each pair of homologous chromosomes, thus ensuring that none of the gametes produced by meiosis will be identical to another. The conceptual and practical consequences of recombination for many aspects of human genetics and genomics are substantial and are outlined in the Box at the end of this section.

Prophase of meiosis I differs in a number of ways from mitotic prophase, with important genetic consequences, because homologous chromosomes need to pair and exchange genetic information. The most critical early stage is called **zygotene,** when homologous chromosomes begin to align along their entire length. The process of meiotic pairing—called **synapsis**—is normally precise, bringing corresponding DNA sequences into alignment along the length of the entire chromosome pair. The paired homologues—now called **bivalents**—are held together by a ribbon-like proteinaceous structure